THE NOVA RESEARCH CURRICULUM

Chemistry

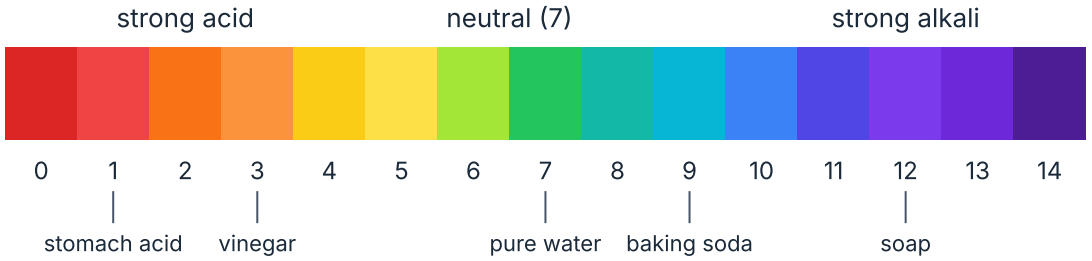
Grade 8 · Term 3

Student edition

Outcome: CH-8.4 acids, bases and salts; pH; neutralisation.

Lesson notes

- **Acids and bases (Weeks 1–4):** acids taste sour and have pH below 7 (lemon juice, vinegar, the hydrochloric acid in the stomach). Bases that dissolve in water are alkalis, with pH above 7 (soap, baking soda, sodium hydroxide). Pure water is neutral (pH 7). Never taste or touch lab chemicals; wear goggles.
- **pH and indicators (Weeks 5–8):** litmus turns red in acid and blue in alkali; universal indicator shows the full pH scale from 0 to 14 by colour. Home indicator: red-cabbage water. Test ten household liquids.



Universal indicator colours on the pH scale.

- **Neutralisation (Weeks 9–12):** acid + base → salt + water; HCl + NaOH → NaCl + H_2O. Practical: add vinegar drop by drop to baking-soda solution with red-cabbage indicator until the colour changes; count the drops (titration-style). Uses: antacids for indigestion, lime on acidic soil, toothpaste against tooth decay.

Workbook pages

Page	Task
1	Sort 12 substances into acid, alkali, neutral.
2	Red-cabbage indicator results table.
3–4	Neutralisation practical report.

Question bank

#	Outcome	Question	Your answer
1	CH-8.4	A pH of 3 is …	neutral / alkaline / acidic / impossible
2	CH-8.4	Litmus turns … in an alkali.	red / blue / green / yellow

#	Outcome	Question	Your answer
3	CH-8.4	Pure water has pH …	0 / 1 / 7 / 14
4	CH-8.4	Acid + base → …	gas only / salt + water / metal + oxygen / acid
5	CH-8.4	Antacid tablets neutralise extra stomach acid. True or false.	True / False
6	CH-8.4	Which is an alkali?	vinegar / lemon juice / soap solution / cola
7	CH-8.4	Name the salt made from HCl and NaOH.	_____
8	CH-8.4	Farmers add lime to soil that is too …	alkaline / acidic / wet / dry